# Imran Jahid Khan

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## Research Interests

Security and Resilience of Cyber-Physical Systems • Sensor Attacks, Anomaly Detection, and Recovery • Robust Reinforcement Learning for Control • Cloud-Controlled and Medical CPS • Blockchain-Based Integrity Verification

## Education

**Tuskegee University, Alabama, USA** *Aug 2025 – May 2027 (expected)*
M.S. in Information Systems and Computer Security
Advisor: Dr. Mohammad Arafatur Rahman
*Relevant Coursework:* Information Security; Network Security and Management; Data Networks & Cloud Computing; Data Mining & Machine Learning; Statistics with R; Data Analytics; Big Data Analytics; Database Systems.

**Bangladesh University of Engineering & Technology (BUET), Dhaka** *2011 – 2016*
B.Sc. in Mechanical Engineering
Thesis: Wind Turbine Blade Design for Optimum Power Output.
*Mathematical Foundations:* Calculus; Linear Algebra; Ordinary & Partial Differential Equations; Complex Variables & Fourier Analysis; Numerical Analysis.

## Research Experience

**Graduate Research Assistant** *Aug 2025 – Present*
Dept. of Computer Science, Tuskegee University • Advisor: Dr. Mohammad Arafatur Rahman

- Built a cloud-controlled Q-learning CartPole testbed; showed that low-variance observation noise interacts non-linearly with environmental stress (inclined surface), cutting mean survival time from 4.82 s to 1.44 s [2].
- Evaluated six DQN training and reconstruction strategies across 240,000 episode records under Gaussian noise and randomized replay; temporal processing (LSTM reconstruction, recurrent DQN) gave the strongest, most consistent replay resilience [7].
- Combined a PPO controller, an LSTM-autoencoder detector, and a private blockchain ledger that detects replay deterministically and anchors recovery to verified clean states; restored 95.7% of clean return on LunarLander, while unverified local-buffer recovery collapsed performance [3].
- Showed that detect-then-recover fails under CGM sensor attacks on an RL artificial pancreas; designed detector-free residual-LSTM reconstruction with a hypoglycemia-averse shield that lowered hypoglycemia risk under replay, drift, and FDI attacks in simglucose [4].
- Co-authored studies of classical vs. quantum RL controllers [5, 8], LSTM-based integrity-attack detection for cloud-based CPS [1], and AI-assisted NMR metabolomics evidence ranking [6].

## Publications

PUBLISHED CONFERENCE PAPERS † presenting author

[1] M. A. Rahman, **I. J. Khan**, and F. Wu, "Detection of Integrity Attacks in Cloud-Based Cyber-Physical Systems (CBCPS) Using Long Short-Term Memory (LSTM) Models," in *Proc. 2026 9th International Conference on Information and Computer Technologies (ICICT)*, pp. 533–538, 2026.

[2] **I. J. Khan**, M. Hossain, M. A. Rahman, and F. Wu, "Resilience Analysis of Cloud-Controlled Cyber-Physical Systems Using a Reinforcement Learning CartPole Testbed," in *Proc. 2026 IEEE World AI IoT Congress (AIIoT)*, pp. 773–779, 2026.

ACCEPTED CONFERENCE / WORKSHOP PAPERS

[3] **I. J. Khan**†, M. Hossain, M. A. Rahman, and F. Wu, "Blockchain-Anchored Detection and Recovery for Resilient Reinforcement Learning-Based Cyber-Physical Systems," *7th Annual National HBCU Blockchain, FinTech, and AI Conference*, National FinTech Center, Morgan State University, 2026.

[4] **I. J. Khan**†, S. Tahmim, M. Hossain, M. A. Rahman, and F. Wu, "Reconstruction-Assisted Resilient Reinforcement Learning for the Artificial Pancreas under CGM Sensor Attacks," *Proc. IEEE International Conference on E-health Networking, Application & Services (HealthCom) Workshops, ISTHS Workshop*, 2026.

[5] M. Hossain, **I. J. Khan**†, M. A. Rahman, and F. Wu, "Evaluating Classical and Quantum Reinforcement Learning Controllers under Meal-Information Uncertainty in Artificial Pancreas Systems," *Proc. IEEE HealthCom Workshops, ISTHS Workshop*, 2026.

[6] S. Tahmim, **I. J. Khan**, and W. E. Collier, "An Auditable Evidence-Ranking Framework for AI-Assisted NMR Metabolomics in Breast Cancer Research," *Proc. IEEE HealthCom Workshops, ISTHS Workshop*, 2026.

SUBMITTED / UNDER REVIEW

[7] **I. J. Khan**, M. Hossain, M. A. Rahman, and F. Wu, "Reconstruction-Assisted DQN for Resilient Cloud-Controlled Cyber-Physical Systems against Data Integrity Attacks," submitted to *2nd Special Session on Cybersecurity and Telecommunication in the Era of AI, 2026 IEEE International Conference on Big Data (BigData)*, 2026.

[8] M. Hossain, **I. J. Khan**, M. A. Rahman, and F. Wu, "Evaluating Learning Efficiency and Computational Trade-Offs in Classical and Quantum Reinforcement Learning Using a CartPole Testbed," submitted to *2026 IEEE International Conference on Big Data (BigData)*, 2026.

## Presentations

- **Oral, Nov 8–10, 2026 (scheduled):** paper [3], 7th Annual National HBCU Blockchain, FinTech, and AI Conference (National FinTech Center, Morgan State University), Nashville, TN.
- **Oral, Oct 2026 (scheduled):** papers [4] and [5], ISTHS Workshop, IEEE HealthCom 2026.
- **Poster, 2025:** "Evaluating Q-Learning Rewards in Noisy Cloud-Based Cyber-Physical Systems." *AWS-MLU AI/ML Fall Teaching and Research Symposium & Student Poster Session*.

## Honors & Scholarships

- **Full Scholarship (Graduate Assistantship)**, M.S. in Information Systems and Computer Security, Tuskegee University, 2025
- **Japan Human Resources Development Scholarship (JDS)**, Government of Japan, for graduate study at Hiroshima University, 2025 *(declined)*
- **Full Scholarship**, M.A. in Policy Economics, Williams College, 2025 *(declined)*
- **Bangladesh Civil Service (Administration) Cadre**, 37th BCS Examination: merit position 89 of 306 appointees from about 243,000 candidates
- **Prime Minister's Fellowship**, Government of Bangladesh, 2024
- **Foreign Master's Scholarship**, "Capacity Building of BCS Administration Academy (2nd Revised)" Project, Government of Bangladesh, 2024 *(declined)*

## Teaching Experience

**Graduate Teaching Assistant** *Aug 2025 – Dec 2025*
Introduction to Computer Science, Dept. of Computer Science, Tuskegee University

## Professional Experience (Bangladesh Civil Service)

**Deputy Director (Senior Assistant Secretary)** *Sep 2024 – Jul 2025*
Export Promotion Bureau, Ministry of Commerce

- Produced data-informed export-policy recommendations and worked with stakeholders to digitize trade-facilitation processes.

**Assistant Commissioner (Land) & Executive Magistrate** *Nov 2022 – Aug 2024*
Upazila Land Office, Patuakhali & Bhola, Ministry of Land

- Managed digital land records and reporting to support land administration; adjudicated land disputes and coordinated field operations.

**Assistant Commissioner & Executive Magistrate** *Apr 2019 – Nov 2022*
Office of the Deputy Commissioner, Cox's Bazar, Ministry of Public Administration

- Coordinated disaster-relief and COVID-19 response using ICT; administered public budgets, personnel, and electronic procurement.

## Technical Skills

**Programming:** Python, C/C++, R, SQL, MATLAB
**ML / RL:** PyTorch; Q-learning, DQN, PPO; LSTM and LSTM autoencoders; anomaly detection; state reconstruction
**Simulation:** Gymnasium (CartPole, LunarLander), simglucose (artificial-pancreas simulator)
**Tools:** Git/GitHub, Linux, LaTeX

## Standardized Tests

**GRE:** 322 (Quantitative 167), Dec 2024 **IELTS Academic:** 7.0, Jan 2024